



# SOC Analyst

## *10 Hands-On Projects for Skills and Resume*

If you're aiming for your first SOC Analyst role, certifications like Security+ or CySA+ are a great start — but hands-on experience is what truly sets you apart. Employers want proof that you can analyze alerts, investigate threats, and document incidents like a real analyst. This guide gives you 10 beginner-to-intermediate level projects that simulate actual SOC tasks.

### Project List

#### **SIEM Log Monitoring & Threat Detection**

Build a Splunk or ELK Stack lab. Ingest logs and detect simulated attacks.

#### **Simulated Phishing Attack & Incident Response**

Recreate a phishing-to-malware scenario and investigate using logs.

#### **IDS Deployment & Packet Analysis**

Install Snort or Suricata and analyze alerts and pcap data with Wireshark.

#### **Cloud Security Monitoring with AWS/Azure Logs**

Enable and analyze cloud logs from CloudTrail or Azure Defender.

#### **Endpoint Monitoring with Sysmon + Wazuh**

Monitor a Windows machine using Sysmon and create detection rules.

#### **Threat Hunting in Real Datasets**

Analyze Windows or Zeek logs from real-world datasets to find hidden threats.

#### **Security Automation Script (SOAR Lite)**

Write a Python/PowerShell script to automate log parsing or IOC enrichment.

#### **Honeypot Deployment & Threat Intel**

Deploy a honeypot and collect/analyze real-world attacker behavior.

#### **Threat Intelligence IOC Research**

Investigate malware/APT campaigns, enrich indicators, and write a threat brief.

#### **Phishing Email & Malware Analysis**

Analyze phishing emails and malicious attachments in a sandboxed environment.

## 1. SIEM Log Monitoring & Threat Detection

**Summary:** Build a Splunk or ELK Stack SIEM lab. Ingest logs (e.g., Sysmon, Linux auth) and detect simulated attacks like brute force or PowerShell misuse.

**Skills:** SIEM use, log parsing, detection engineering, correlation.

**Tools:** Splunk Free, ELK Stack, Sysmon, Windows/Linux VMs.

**Time:** ~20–30 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★★★★★ (10/10)

### What it simulates:

Real Tier 1–2 SOC alert triage, log analysis, and use case development.

Hands-on detection engineering in a monitored environment.

### Add to Resume:

- “Built Splunk SIEM lab to ingest and analyze Sysmon/Windows logs; created custom detection rules for brute force and PowerShell-based attacks.”
- “Simulated real-world attacks and investigated events using ELK Stack, improving response skills and detection logic.”

## 2. Simulated Phishing Attack & Incident Response

**Summary:** Simulate a phishing-to-malware scenario in a lab. Investigate with logs and produce a detailed incident response report.

**Skills:** IR lifecycle, Windows log analysis, evidence collection, reporting.

**Tools:** Windows VM, Sysinternals, Event Viewer, Kali, SET.

**Time:** ~15–25 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★★★★★ (9/10)

### What it simulates:

A full SOC incident involving phishing, malware, log review, containment, and documentation.

Simulates Tier 1–2 escalation and response workflow.

### Add to Resume:

- “Simulated phishing attack in lab and performed full IR process including log review, malware identification, and executive-level reporting.”
- “Traced infection path via Windows event logs (Event ID 4688, Sysmon logs) and documented remediation steps in formal IR report.”



### 3. IDS Deployment & Packet Analysis

**Summary:** Deploy Snort or Suricata. Simulate attacks (e.g., Nmap, SQLi) and analyze alerts and PCAPs using Wireshark.

**Skills:** IDS/IPS, network protocols, rule tuning, packet analysis.

**Tools:** Snort, Suricata, Wireshark, Metasploitable, Kali.

**Time:** ~15–20 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★★★★★ (9/10)

**What it simulates:**

Real-time network intrusion detection tasks.

Alert investigation based on IDS telemetry—common in Tier 1–2 SOC environments.

**Add to Resume:**

- “Deployed Suricata IDS to monitor test network; generated attack traffic and tuned rules to reduce false positives and detect SQL injection attempts.”
- “Investigated PCAPs using Wireshark and correlated alerts with MITRE ATT&CK TTPs.”

### 4. Cloud Security Monitoring with AWS/Azure Logs

**Summary:** Enable CloudTrail or Defender logs. Simulate events (e.g., IAM abuse, unauthorized S3 access) and analyze them via console or SIEM.

**Skills:** Cloud security, logging, IAM misconfig detection.

**Tools:** AWS Free Tier, GuardDuty, CloudTrail, ELK/Splunk.

**Time:** ~10–20 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★★★★★ (9/10)

**What it simulates:**

Tier 1/2 monitoring of cloud security logs in modern hybrid SOCs.

Detecting IAM privilege escalation, misconfigurations, cloud anomalies.

**Add to Resume:**

- “Configured AWS CloudTrail/GuardDuty to detect and alert on suspicious IAM events including unauthorized access and misconfigured S3 buckets.”
- “Analyzed cloud logs and set up alerting rules to simulate SOC Tier 2 cloud monitoring workflows.”



## 5. Endpoint Monitoring with Sysmon + Wazuh

**Summary:** Use Sysmon and Wazuh for endpoint log collection. Simulate malware behavior and trigger alerts on suspicious actions.

**Skills:** Host-based detection, event ID mapping, Sigma/YARA rules.

**Tools:** Sysmon, Wazuh, Windows VM, PowerShell.

**Time:** ~10–15 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★★★ (8/10)

### What it simulates:

Tier 1–2 endpoint alert investigation in EDR-like setups (e.g., CrowdStrike, Defender ATP).

Detecting suspicious behavior via host logs (e.g., credential dumping, persistence).

### Add to Resume:

- “Ingested Sysmon logs into Wazuh and developed custom alert rules to detect credential dumping and persistence tactics.”
- “Simulated malware behavior and validated endpoint alerts via process tree and registry activity.”

## 6. Threat Hunting in Real Datasets

**Summary:** Analyze log datasets (Windows, Zeek) for anomalies. Build a timeline of attacker activity based on raw logs.

**Skills:** SPL/KQL, anomaly detection, ATT&CK mapping.

**Tools:** Splunk, Elastic, CyberDefenders datasets.

**Time:** ~10–20 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★★★ (8/10)

### What it simulates:

Tier 2 threat hunting workflow: identify suspicious patterns missed by alerts.

Investigation using structured queries and hypothesis-driven analysis.

### Add to Resume:

- “Performed threat hunt on Zeek and Windows log dataset; identified lateral movement and brute-force login indicators using SPL queries.”
- “Mapped findings to MITRE ATT&CK framework and built a comprehensive attack timeline.”



## 7. Security Automation Script (SOAR Lite)

**Summary:** Build a Python or PowerShell script to enrich IOCs, parse logs, or auto-tag suspicious events.

**Skills:** Scripting, API use, automation logic.

**Tools:** Python, VirusTotal API, AbuseIPDB, log samples.

**Time:** ~8–15 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★ (7/10)

### What it simulates:

Automating Tier 1 enrichment and triage tasks seen in SOAR platforms.

Saves analyst time by streamlining manual processes.

### Add to Resume:

- “Developed Python script to enrich IOC indicators via VirusTotal/AbuseIPDB APIs and summarize alert context in JSON reports.”
- “Automated noisy log filtering using PowerShell to support faster SOC response.”

## 8. Honeypot Deployment & Threat Intel

**Summary:** Set up Cowrie SSH honeypot to capture attacker activity. Analyze login attempts, files, and attacker IPs.

**Skills:** Real attacker interaction, log analysis, OSINT enrichment.

**Tools:** Cowrie, ELK, VirusTotal, DigitalOcean.

**Time:** ~15 hrs over 1–2 weeks

**Cost:** \$0–\$10

**Resume Value:** ★★★★★★ (7/10)

### What it simulates:

External threat intelligence monitoring, IOC extraction, and attack analysis.

Observing real brute-force and botnet behavior.

### Add to Resume:

- “Deployed SSH honeypot in cloud VM; captured and analyzed 5,000+ attacker attempts including common TTPs and password combos.”
- “Enriched IPs with OSINT tools and documented findings in threat brief.”

## 9. Threat Intelligence IOC Research

**Summary:** Research malware/APT, collect IOCs, enrich with tools (e.g., VirusTotal), and build a threat profile.

**Skills:** IOC research, enrichment, ATT&CK mapping, reporting.

**Tools:** VirusTotal, OTX, WHOIS, MISP.

**Time:** ~10–15 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★ (6/10)

### What it simulates:

SOC threat intelligence feed handling, watchlist creation, adversary profiling.

Assists Tier 2–3 analysts with threat correlation and prevention.

### Add to Resume:

- “Compiled threat report on Emotet malware campaign with enriched IOCs and MITRE TTP mapping; integrated with open-source MISP platform.”
- “Correlated indicators across logs and feeds to identify targeted activity.”

## 10. Phishing Email & Malware Analysis

**Summary:** Analyze phishing email headers, malicious links, and malware attachments in a sandbox. Extract and report IOCs.

**Skills:** Email forensics, malware sandboxing, IOC extraction.

**Tools:** Any.Run, VirtualBox, VirusTotal, Wireshark.

**Time:** ~6–12 hrs

**Cost:** \$0

**Resume Value:** ★★★★★★ (6/10)

### What it simulates:

Tier 1 SOC phishing response—analyzing user-reported suspicious emails.

Triage of email threats and basic malware behavior analysis.

### Add to Resume:

- “Analyzed phishing email with malicious doc attachment; performed sandbox detonation and extracted PowerShell-based downloader behavior.”
- “Mapped attack to initial access vector and blocked indicators at firewall and EDR levels.”